

LECTURE 14: MATH1061/7861
TOPICS: SEQUENCES, MATHEMATICAL INDUCTION
DATE: WEDNESDAY MARCH 25, 2026

Learning objectives:

- (1) Gain fluency in the notation used for sequences and summations.
- (2) Use mathematical induction to prove statements.

* MATH 7861 Assignment is on the course website.

* Next week, there are no ~~Applied~~ Applied classes on Friday (Easter). So you have to go to earlier classes.

$$5^k + 9 < 6^k$$

multiply $\times 5$

$$\Rightarrow 5(5^k + 9) < 5 \cdot 6^k$$

$$\Rightarrow \underline{5^{k+1} + 45} < 5 \cdot 6^k$$

Q1: Prove using induction that

$$\forall n \in \mathbb{Z}_{\geq 2}, \quad 5^n + 9 < 6^n$$

• Base Case: Take $n=2$. Then we have

$$5^2 + 9 = 25 + 9 = 34 < 36 = 6^2.$$

• Inductive hypothesis: Suppose statement is true for some $n = k \geq 2$; i.e. $5^k + 9 < 6^k$ *.

• What we want to prove: The statement for $n = k+1$; i.e. want to show $5^{k+1} + 9 < 6^{k+1}$

• Inductive step:

$$5^{k+1} + 9 < 5^{k+1} + 45 = 5(5^k + 9)$$

$$\begin{aligned} \left[\text{By inductive hypothesis} \right] &< 5 \cdot 6^k \\ &< 6 \cdot 6^k = 6^{k+1} \end{aligned}$$

Q2: Define a sequence as follows:

$$a_1 = 1, \quad a_{n+1} = a_n + 2^n \quad \forall n \geq 1$$

(a) Write down the values of a_2, a_3, a_4, a_5 .

$$a_2 = a_1 + 2^1 = 1 + 2 = 3 = 4 - 1 = 2^2 - 1$$

$$a_3 = a_2 + 2^2 = 3 + 4 = 7 = 8 - 1 = 2^3 - 1$$

$$a_4 = a_3 + 2^3 = 7 + 8 = 15 = 16 - 1 = 2^4 - 1$$

$$a_5 = a_4 + 2^4 = 15 + 16 = 31 = 32 - 1 = 2^5 - 1$$

(b) Guess what a_n is.

$$a_n = 2^n - 1 \quad \forall n \geq 1.$$

(c) Prove your guess using induction.

Base Case: $n=1$ $a_1 = 1 = 2^1 - 1$

Inductive hypothesis: Statement is true for $n=k$;
i.e., $a_k = 2^k - 1$.
what we want to prove: $a_{k+1} = 2^{k+1} - 1$

Inductive step:

$$a_{k+1} = a_k + 2^k$$

By definition

$$= (2^k - 1) + 2^k$$

By inductive hypothesis

$$= 2^k + 2^k - 1 = 2 \times 2^k - 1 = 2^{k+1} - 1.$$

Q3: For each $n \in \mathbb{Z}_{\geq 1}$, define

$$a_n := \sum_{i=1}^n (2i-1)$$

(1) Write down a_1 , a_2 , a_3 , and a_4 .

$$a_1 = \sum_{i=1}^1 (2i-1) = (2 \times 1 - 1) = 1$$

$$a_2 = \sum_{i=1}^2 (2i-1) = (2 \times 1 - 1) + (2 \times 2 - 1) = 1 + 3 = 4$$

$$a_3 = \sum_{i=1}^3 (2i-1) = (2 \times 1 - 1) + (2 \times 2 - 1) + (2 \times 3 - 1) = 1 + 3 + 5 = 9$$

$$a_4 = \sum_{i=1}^4 (2i-1) = (2 \times 1 - 1) + (2 \times 2 - 1) + (2 \times 3 - 1) + (2 \times 4 - 1) = 1 + 3 + 5 + 7 = 16$$

(2) Find a general formula for a_n .

Guess: $a_n = n^2$

(3) Use induction to prove your formula is correct.

We want to prove that $a_n = n^2$ for all $n \in \mathbb{Z}_{\geq 1}$.

Base case: Take $n=1$. Then $a_1 = 1 = 1^2$, so the statement holds for $n=1$.

Inductive hypothesis: Suppose the statement is true for $n=k \geq 1$.

I.e. $a_k = k^2$.

We want to show that $a_{k+1} = (k+1)^2$.

Inductive step.

From the definition of the sequence, $a_{k+1} = \sum_{i=1}^{k+1} (2i-1)$, so

$$a_{k+1} = \sum_{i=1}^{k+1} (2i-1) = (2(k+1)-1) + \sum_{i=1}^k (2i-1)$$

$$= (2(k+1)-1) + a_k$$

$$= (2(k+1)-1) + k^2 \quad \left. \begin{array}{l} \text{via the} \\ \text{inductive hypothesis} \end{array} \right\}$$

$$= 2k + 2 - 1 + k^2$$

$$= k^2 + 2k + 1 = (k+1)^2$$

□